

MSK MODULE OBJECTIVES				
Activity	Discipline	Title	OBJECTIVE CODE	Lecture Objective
DLA 01	ANAT	Introduction to Arteries, Veins and Lymphatics	SOM.MK.II.BPM1.1.MSK.1.ANAT.1037	List the components of the cardiovascular system
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1038	Describe the basic structure and function of arteries, veins and lymphatic vessels
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1039	Describe the flow of blood in the cardiovascular system
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1040	Describe the basic structure and function of lymph nodes
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1041	Describe the flow of lymph
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1042	Describe the terms anastomoses and collateral circulation
DLA 02	ANAT	Somatic Nerves and Plexuses	SOM.MK.II.BPM1.1.MSK.1.ANAT.1064	Define the terms somatic afferent, visceral afferent, somatic efferent, and visceral efferent as they apply to nerve fibers found in spinal nerves.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1065	List the different modalities and targets of the fibers found in spinal nerve
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1066	Compare and contrast differences between dorsal and ventral roots, compare the contrast differences between dorsal and ventral rami.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1067	Describe a dorsal root ganglion and the pathway of a typical afferent neuron.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1068	Define a dermatome, myotome and discuss the clinical importance of understanding these concepts.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1069	Describe which dermatomes are associated with the following anatomical landmarks/structures: lateral border of upper limb, thumb, medial border of upper limb, little finger, nipple, xiphoid, umbilicus, pubic region, big toe, little toe.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1070	Explain how the spinal nerves exit the five regions of the vertebral column and use this information to interpret clinical signs and symptoms
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1071	Describe the organization and composition of a somatic nerve plexus and list the nerve fiber types found in each component.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1072	Outline the dermatome map of the body and use it to explain the reappearance of herpes zoster (shingles) in a particular dermatome(s).
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1073	Describe the pathways of sympathetic fibers innervating the body wall
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1074	Describe the course of visceral sensory fibers traveling with sympathetic nerves, and the location of their cell bodies.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1075	Describe the mechanism of referred pain.
DLA 03	ANAT	Vertebral Column and Spinal Cord	SOM.MK.II.BPM1.1.MSK.1.ANAT.1043	Describe the different components of a typical vertebra, highlighting the features that are unique to each region.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1044	Describe the joints found between and associated movements of the typical vertebrae.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1045	Describe the structure of the atlas and the axis.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1046	Describe the structure of the intervertebral disc.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1047	Describe the joints between the atlas, axis and skull, including the movements possible at each.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1048	Describe the function of the transverse and alar ligaments of the atlas.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1049	Describe the five ligaments that support the vertebral column.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1050	Describe the normal curvatures of the vertebral column and their development.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1051	Describe scoliosis, kyphosis and lordosis
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1052	Describe the difference between spinal gray matter and spinal white matter with respect to location, composition and appearance (Review from FTM)
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1053	Describe the external shape of the spinal cord and the reason for its different appearance in some locations (Review from FTM)
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1054	Describe the organization and appearance of gray matter in the different regions of the spinal cord (Review from FTM)
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1055	Describe the functional modalities associated with the ventral, dorsal and lateral gray horns (Review from FTM)
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1056	Describe the extent, attachments and spaces of the spinal meninges.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1057	Describe the extent and the contents of the dural sac (lumbar cistern).
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1058	Describe the extent and the attachments of filum terminale and denticulate ligaments.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1059	Describe the changes in vertebral level of the conus medullaris from 8 weeks to 24 weeks to newborn to adult.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1060	Describe at which vertebral level a lumbar puncture (spinal tap) should be performed in an infant/child versus in an adult, and why.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1061	List the structures the needle would pierce in a lumbar puncture.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1062	Describe the location and function of the internal vertebral venous plexus (of Batson) and its role in the spread of infections and/or cancer.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1063	Describe the blood supply and venous drainage of the spinal cord and its clinical significance.
Lecture 01	ANAT	The Back	SOM.MK.II.BPM1.1.MSK.1.ANAT.1076	Describe the structure of the scapula and clavicle, and their function in attachment of the appendicular (shoulder) to the axial skeleton (trunk).
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1077	Describe the general difference in function between the superficial (extrinsic) and deep (intrinsic) muscles of the back.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1078	Describe the general attachments, innervations, functions, major blood supply, venous drainage, and lymphatic drainage of the superficial and deep muscles of the back.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1079	Describe the general extent of the thoracolumbar fascia and how it encloses the intrinsic muscles of the back
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1080	Describe the general attachments, innervations, functions and major blood supply of the erector spinae muscles.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1081	Describe the general location, innervations and actions of the transversospinalis muscles.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1082	Describe the differences in action between unilateral and bilateral contraction of the muscles of the back.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1083	Describe the cutaneous innervation of the back.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1084	Describe the location of the lymph nodes into which the lymphatics of the skin of the back drain.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1085	List the borders of the triangle of auscultation and define its clinical significance.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1086	Describe the triangle of Petit and its clinical significance.
Lecture 02	ANAT	Introduction to Clinical Imaging and Imaging of Back	SOM.MK.II.BPM1.1.MSK.1.ANAT.1032	Discuss the following types of medical imaging: Radiographs, CT scans, MRI and contrast studies
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1112	Identify the major bony landmarks of 5 regions of vertebral column in imaging studies and use this information to interpret clinical signs and symptoms
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1107	Identify the superficial and deep muscles of the back in imaging studies

			SOM.MK.II.BPM1.1.MSK.1.ANAT.1111	Describe the major anatomical features of spinal cord, nerves and meninges in imaging studies.
Lecture 03	ANAT	Clinical Anatomy of the Back	SOM.MK.II.BPM1.1.MSK.1.ANAT.1108	Describe the anatomical location and mechanism of hyperextension injuries of the neck.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1121	Describe and identify (in imaging studies) the anatomical location, mechanism and clinical sequences of a Jefferson fracture, compression fractures of the vertebrae including wedge and Hangman's fracture
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1114	Describe the anatomical location, mechanism and clinical sequences of fractures of the dens and identify them in imaging studies
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1109	Describe and identify brevicollis (Klippel-Feil) and hemivertebra and explain how they can lead to scoliosis (Grey's Anatomy For Students "In the Clinic" Blue Boxes)
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1110	Describe sacralization of the lumbar vertebrae and lumbarization of the sacral vertebrae (Grey's Anatomy For Students "In the Clinic" Blue Boxes)
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1115	Describe the process of disc herniation and its consequences in terms of nerve impingements paying particular attention to the differences between the cervical, thoracic and lumbar regions.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1116	Identify herniation using radiographic studies and use this information to interpret clinical signs and symptoms
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1117	Describe the anatomical sequelae after rupturing the transverse ligament of the atlas.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1118	Describe the anatomical landmarks used for lumbar epidural and sacral (caudal) epidural anesthesia and explain when each will be performed.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1120	Describe the mechanism(s) that produces lumbar spinal stenosis and use this information to interpret clinical signs and symptoms (SG Article)
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1119	List the neurological signs and symptoms of spondylolysis and spondylolisthesis and be able to identify them in imaging studies.
DLA 04	ANAT	Development of the Spinal Cord and Vertebrae	SOM.MK.II.BPM1.1.MSK.1.ANAT.1094	Describe the formation of the neural plate and define the term neuroectoderm
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1095	Describe the major steps in formation of the neural tube and neural crest.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1096	List the derivatives of the neural tube.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1097	List the nervous system derivatives of the neural crest cells and briefly list other tissues or organs are formed from neural crest cells.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1098	Describe the three zones which form in the neural tube and explain how each of these zones contributes to formation of the spinal cord.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1099	Describe the anatomical sites of formation of somatic and visceral motor, and sensory, neurons from neuroblasts.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1100	Describe the development of the meninges.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1101	Describe the embryological origin of the vertebrae and intervertebral discs.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1102	Describe the location of the primary and secondary ossification centers of a typical vertebra and give the time at which they occur.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1103	List common causes of neural tube defects.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1104	Describe the embryological processes that fail and subsequent anatomical defect and severity of; anatomy and severity of spina bifida occulta, spina bifida cystica with meningocele, spina bifida cystica with meningocele and spina bifida with myeloschisis.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1105	Describe the embryological formation and clinical presentation of chordoma.
Lecture 04	ANAT	Clinical Anatomy of Axilla and Brachial Plexus	SOM.MK.II.BPM1.1.MSK.1.ANAT.1122	Describe the location of the axilla and the structures that form its medial, lateral, anterior and posterior boundaries (walls), and its apex.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1123	Describe the cervicoaxillary sheath and its clinical importance.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1124	Describe the clinical significance of the axillary sheath in local anesthesia.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1125	Describe the subclavian artery and its branches, based on what they supply and where they are located.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1126	List the general attachments, innervations and actions of the muscles that contribute to the walls of the axilla.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1127	Describe the functional loss of the muscles of the axillary walls resulting from lesions to the nerves innervating these muscles.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1128	Describe the formation of the axillary vein and the major tributaries that join it.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1129	Describe the division of the axillary artery into its three parts and the structures related to each part.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1501	Describe the boundaries and the contents of the subscapular spaces including the quadrangular space, the triangular space and the triangular interval
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1130	List the arteries involved in anastomosis around the scapula, give the origin artery for each and discuss the clinical importance of this anastomoses.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1131	Describe the lymphatic drainage of the walls of the axilla use this information to interpret clinical signs and symptoms
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1132	Describe the organization of the brachial plexus from its roots to its terminal branches.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1133	Define the terms pre-fixed and post-fixed brachial plexus and discuss its clinical importance.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1134	List the fiber types of the nerves that are present in the branches of the brachial plexus.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1135	Describe the anatomical relationships that are important for successful insertion of a catheter for central venous access (infraclavicular subclavian venipuncture) (Grey's Anatomy For Students "In the Clinic" Blue Boxes)
DLA 05	ANAT	Clinical Anatomy of Shoulder	SOM.MK.II.BPM1.1.MSK.1.ANAT.1136	Describe the movements that occur at the shoulder joint
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1137	Identify, name and orient the bones forming the shoulder girdle and the shoulder joint.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1138	Describe the 3 joints that form the shoulder complex
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1139	Describe the location of the subacromial and subscapular bursae at the shoulder joint.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1140	Describe the ligamentous injuries and clinical signs in acromioclavicular (shoulder) separation and identify this on imaging studies.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1141	Describe the signs and symptoms of clavicular fracture, and how the fragments of the fracture would be displaced
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1142	List the major elements that contribute to the stability of the glenohumeral joint.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1144	Explain why the glenohumeral joint can be easily dislocated.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1145	Describe shoulder dislocation injuries, particularly antero - inferior dislocations, and explain the nerve injuries that might accompany shoulder dislocations.

			SOM.MK.II.BPM1.1.MSK.1.ANAT.1143	Describe the rotator cuff muscles and their function in maintaining the stability of the glenohumeral joint.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1147	Describe the anatomical and clinical signs of rotator cuff disorders due to injury of the supraspinatus muscle
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1148	Describe the cutaneous innervation of the shoulder.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1149	List the attachments, innervations and actions of the muscles that act on the shoulder joint.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1503	Describe the location and boundaries of the suprascapular foramen and its contents
Lecture 05	ANAT	Clinical Anatomy of the Arm and Forearm	SOM.MK.II.BPM1.1.MSK.1.ANAT.1150	Describe cubital tunnel syndrome and the clinical consequences (SG Article)
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1151	List the general attachments, innervations, blood supply and actions of the muscles in the flexor and extensor compartments of the arm
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1152	Describe the cutaneous innervation of the arm
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1153	Describe the course of the brachial artery and its branches.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1156	Describe the movements that occur at the elbow joint
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1157	Describe the bony and ligamentous anatomy of the elbow joint.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1158	Describe the movements that occur at the proximal and distal radioulnar joints
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1159	Describe the arterial anastomosis of the elbow.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1160	Describe the borders of the cubital fossa, its contents and their relationships.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1161	Describe the anatomical location, mechanism and clinical sequences of elbow joint dislocation
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1163	Describe the cause and signs of Volkmann's ischemic contracture.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1164	Describe the basic mechanism of radial head subluxation ("Pulled elbow") in small children and identify it on a radiograph
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1165	Describe and identify fracture of head of radius and possible clinical consequences.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1166	Describe the common flexor and common extensor tendons and the signs and symptoms of inflammation of these tendons
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1167	List the general attachments, innervations, blood supply and actions of the muscles in the flexor and extensor compartments of the forearm
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1168	Describe the cutaneous innervation of the forearm
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1169	Describe the course of the radial and ulnar arteries in the forearm
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1170	Describe the landmarks used when taking the radial pulse (SG)
Lecture 06	ANAT	Clinical Anatomy of the Wrist and Hand	SOM.MK.II.BPM1.1.MSK.1.ANAT.1174	Describe the movements that occur at the wrist joint
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1175	Describe the bony and ligamentous anatomy of the wrist joint.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1176	Describe the carpal tunnel and Guyon's canal
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1177	Describe the articulations between the distal radius and the proximal row of carpal bones.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1178	Describe the general arrangement between the distal row of carpal bones and the metacarpals (4 distal carpals, 5 metacarpals).
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1179	List the most commonly fractured carpal bone
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1180	List the most commonly dislocated bone in the carpus and how it can lead to carpal tunnel syndrome
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1181	Describe and identify Boxers fractures and possible clinical consequences.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1182	Identify the following carpal bone fractures: Scaphoid fracture and (Fracture of hook of hamate) and describe their possible clinical consequences.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1183	Describe the fascia of the wrist and hand including the flexor and extensor retinaculum, palmar carpal ligament and palmar aponeurosis
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1184	Describe the fibrous digital tendon sheaths and their function in preventing bowing of tendons
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1185	Describe the anatomical and clinical mechanism of De Quervains syndrome (Grey's Anatomy For Students "In the Clinic" Blue Boxes)
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1186	Describe the formation and function of the extensor hoods
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1187	Describe the synovial sheaths in the hand and fingers and their importance in the spread of infections.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1188	Describe the presentation of Dupuytren's contracture and "trigger finger" (stenosing tenosynovitis).
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1189	List the attachments, innervations and actions of the intrinsic muscles of the hand.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1190	Describe the movements that occur at the carpometacarpal, metacarpophalangeal and interphalangeal joints of the thumb and of digits 2-5.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1191	Discuss the anatomical basis for Allen's test, paying specific attention to the origin of deep and superficial palmar arches.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1192	Describe the boundaries and contents of the "anatomical snuff box".
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1193	Describe the cutaneous innervation of the hand
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1194	Describe the dermatomes of the upper limb
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1197	Describe the venous and lymphatic drainage of the arm, forearm and hand.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1198	List the sites commonly used for venipuncture in the upper limb (Grey's Anatomy For Students "In the Clinic" Blue Boxes)
Lecture 07	ANAT	Imaging of the Upper Limb	SOM.MK.II.BPM1.1.MSK.1.ANAT.1155	Identify the shoulder joint in imaging studies including but not limited to CT, MRI and radiographic images.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1146	Identify antero - inferior shoulder dislocation on a radiograph.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1154	Identify the bones of the arm in imaging studies including and use this information to interpret clinical signs and symptoms.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1162	Explain the potential clinical consequences of fractures at the surgical neck, midshaft, supracondylar, and medial epicondylar regions of the humerus and be able to recognize them on a radiograph.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1172	Identify the elbow joint in imaging studies and use this information to interpret clinical signs and symptoms.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1171	Identify the bones of the forearm in imaging studies including and use this information to interpret clinical signs and symptoms.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1173	Recognize and describe Colle's fracture and Smith's fracture on a radiograph.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1196	Identify the wrist and hand joints in imaging studies including but not limited to CT, MRI and radiographic images.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1195	Identify the bones of the hand in imaging studies including but not limited to CT, MRI and radiographic images.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1199	Describe the course of the radial, median, musculocutaneous, axillary and ulnar nerves in the arm, forearm and hand
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1200	Describe the signs and symptoms of an upper brachial plexus nerve lesion(s).
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1201	Describe the signs and symptoms of an lower brachial plexus nerve lesion(s).

Lecture 08	ANAT	Flipped Class: Upper Limb Clinical Cases	SOM.MK.II.BPM1.1.MSK.1.ANAT.1202	Explain how a cervical rib may contribute to the development of a lower brachial plexus lesion
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1203	Describe the motor and sensory deficits of lesions of the radial nerve in the axilla, arm and wrist.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1204	Explain why radial nerve injury at/above the elbow decreases grip strength.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1205	Describe the signs and symptoms of lesions of the median nerve in the arm, wrist and proximal palm.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1206	Explain the anatomical mechanism, signs and symptoms of “carpal tunnel syndrome” and the special test used for diagnosis.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1208	Describe the signs and symptoms of lesions of the ulnar nerve in the arm and wrist and palm.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1162	Explain the potential clinical consequences of fractures at the surgical neck, midshaft, supracondylar, and medial epicondylar regions of the humerus and be able to recognize them on a radiograph.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1182	Identify the following carpal bone fractures: scaphoid fracture and fracture of hook of hamate and describe their possible clinical consequences.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1210	Describe the functional tests for integrity of the median, ulnar and radial nerves in the forearm and hand (SG)
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1211	Describe the biceps, triceps and brachialis reflexes and list the nerve roots being tested (SG)
DLA 06	ANAT	Thoracic Wall and Pectoral Region	SOM.MK.II.BPM1.1.MSK.1.ANAT.1212	Describe the skeletal framework of the thoracic wall.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1213	Describe the costovertebral, sternocostal and sternoclavicular joints.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1214	List the general attachments, innervations and actions of the intercostal muscles.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1215	Describe the course of the intercostal neurovascular bundle
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1216	List the innervations, blood supply and actions of the muscles of the anterior and lateral chest wall.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1217	Discuss and describe the the bucket-handle and pump-handle movements of the ribs during respiration.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1218	Describe the cutaneous innervation and dermatomes of the thoracic wall.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1219	Describe the boundaries of the superior and inferior thoracic apertures.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1220	List the layers of the thoracic wall from skin to the parietal pleura.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1221	Describe the locations of the parietal pleura and its reflection lines.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1222	List the recesses formed by the pleural reflections.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1223	Describe the arterial supply, venous drainage and innervation of the parietal pleura.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1224	Describe the arterial supply, venous and lymphatic drainage of the thoracic wall.
Lecture 09	ANAT	Imaging of the Thorax	SOM.MK.II.BPM1.1.MSK.1.ANAT.1235	Identify the vertebrae, ribs, sternum, clavicle, anterior and posterior thoracic wall, on imaging studies.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1236	Identify the muscles of the thoracic wall on imaging studies.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1237	Identify rib fractures on imaging studies.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1238	Describe flail chest, its mechanism of action and use this information to interpret clinical(radiographic) signs and symptoms.
Lecture 10	ANAT	Clinical Anatomy of the Thorax	SOM.MK.II.BPM1.1.MSK.1.ANAT.1226	Explain the relationship between dermatomes of the thoracic wall and the emergence/appearance of herpes zoster (shingles).
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1227	Describe the “milk line” and explain its clinical significance.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1228	Describe polythelia, polymastia, amastia, gynecomastia and their clinical significance.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1229	Explain the clinical importance of the lymphatic drainage of a cancerous mammary gland.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1230	Describe the anatomy and the clinical importance of the retro mammary space and the axillary tail of Spence.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1231	Describe the anatomical basis for the visible signs of breast cancer including peau d’orange, skin dimpling and nipple retraction.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1232	Describe a sentinel lymph node and its clinical significance.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1233	Describe the neurovascular structures that are at risk during a radical mastectomy.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1234	Describe lymphedema and how it can be a consequence of radical mastectomy with lymphatic clearance.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1239	Define pleurisy (pleuritis) and explain why it causes localized (somatic) pain
Lecture 11	ANAT	Anterior Abdominal Wall and Inguinal Region	SOM.MK.II.BPM1.1.MSK.1.ANAT.1240	Describe thoracocentesis and explain why the needle is passed in the lower border of the intercostal space
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1241	Describe the ribs levels at the midclavicular, midaxillary and paravertebral lines where a thoracocentesis can be performed
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1242	Describe the bony boundaries of the anterolateral abdominal wall.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1243	List the dermatomes which overlie the xiphoid process, umbilicus and pubis.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1244	Describe the division of the anterolateral abdominal wall into four clinical quadrants and nine anatomical regions and its clinical significance.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1245	Describe the location of the major abdominal organs and vessels in the four quadrants and in the nine regions of the abdomen.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1246	Describe the blood supply, venous drainage, lymphatic drainage and cutaneous innervation of the abdominal wall.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1247	List the layers of the superficial abdominal fascia and describe their locations.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1248	List the general attachment, blood supply, innervations, and function of the muscles of the anterolateral abdominal wall.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1249	Describe how the aponeuroses of these muscles contribute to the anterior and posterior layers of the rectus sheath.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1250	Describe the inguinal ligament and its attachment sites.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1251	Describe the anatomy of the superficial and deep inguinal rings.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1252	Describe the boundaries of the inguinal triangle (of Hesselbach) and its clinical significance.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1253	Explain the difference between a direct and an indirect inguinal hernia.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1254	Explain why pain from an inguinal hernia occurs at the site of the hernia.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1255	Briefly describe testicular descent and use this information to interpret clinical signs and symptoms that may be associated with cryptorchidism and ectopic testis.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1256	List the layers of spermatic fascia and where they arise from
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1257	Describe the fate of the processus vaginalis and its contribution to an indirect inguinal hernia.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1258	Describe the formation of hydrocele

			SOM.MK.II.BPM1.1.MSK.1.ANAT.1259	Explain the surgical significance of the intersection of the right linea semilunaris and the costal margin.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1260	Explain the surgical significance of "McBurney's point".
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1261	Explain the surgical approach through the layers of the abdominal wall to approach an inflamed appendix and the surgical significance of the muscle fiber orientation.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1262	Explain the advantage of performing vertical surgical incisions in the midline.
Lecture 12	ANAT	Posterior Abdominal Wall	SOM.MK.II.BPM1.1.MSK.1.ANAT.1263	Describe the boundaries of the posterior abdominal wall.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1264	Describe the attachment, innervation and blood supply of the diaphragm.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1265	Describe the attachment sites and the peritoneal lining of the diaphragm.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1266	List the innervations, blood supply, general attachments and actions of the muscles of the posterior abdominal wall.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1267	Describe the origin of the lumbar plexus and list its branches and their distributions.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1270	Describe the inferior vena cava and its tributaries, and list the vertebral level where the common iliac veins join.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1271	Describe the cisterna chyli as well as the thoracic duct and use this information to interpret clinical signs and symptoms.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1502	Explain how venous blood may reach the right atrium when the IVC is obstructed
Lecture 13	ANAT	Pelvis	SOM.MK.II.BPM1.1.MSK.1.ANAT.1272	List the bones of the pelvic girdle and name the components of the individual bones that comprise the os coxa.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1273	Identify the bony landmarks of the pelvis which can be palpated on the body surface or palpated per vagina or rectum.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1274	Describe the sacroiliac joints and their ligamentous support.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1275	Explain the importance of the sacrospinous and sacrotuberous ligaments in preventing rotation of the sacrum at the sacroiliac joints.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1276	List the openings in the bony pelvis, major structures passing through these openings, and with which spaces these openings communicate.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1277	Describe the major differences between the male and female bony pelvis.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1278	Describe the following dimensions of the pelvis: pelvic inlet, conjugate (antero-posterior) diameter, interspinous distance and pelvic outlet.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1279	Describe the boundaries of the "true" and "false" pelvis.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1280	Describe the general attachments, innervations and actions of the muscles of the pelvic wall.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1281	List the muscles that make up the pelvic diaphragm.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1282	List the divisions, attachments, innervations, blood supply and general function of the levator ani muscle.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1283	Describe the urogenital hiatus in the pelvic floor and list the major structures passing through the hiatus in the male and female
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1284	Describe the areas of supply of the branches of the internal iliac artery in males and females.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1285	Describe the function of levator ani in maintaining urinary and fecal continence.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1286	Describe the course of the obturator nerve, artery and vein.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1287	List the ventral rami contributing to the sacral plexus and describe the relationship of this plexus to the piriformis muscle.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1288	List the terminal branches of the sacral plexus and their distribution
Lecture 14	ANAT	Imaging of the Abdomen and Pelvis	SOM.MK.II.BPM1.1.MSK.1.ANAT.1268	List the parietal and visceral branches of the abdominal aorta and use this information to identify them on cadaveric specimen and imaging studies.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1269	List the approximate vertebral levels of the visceral branches of the abdominal aorta and its bifurcation into the common iliac arteries and their distribution.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1289	Identify the bones, muscles and vessels of the anterior/posterior abdominal wall and pelvic wall on imaging studies.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1495	Describe types of pelvic fractures based on mechanism of injury
Lecture 15	ANAT	Clinical Anatomy of the Lower Limb - Gluteal Region, Hip Joint, Thigh	SOM.MK.II.BPM1.1.MSK.1.ANAT.1290	Describe the bones, including their major features that contribute to the formation of the lower limb.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1291	Describe the line of weight (aka center of gravity) in relationship to the hip, knee and ankle joints.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1292	Describe the stages of gait
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1293	Describe the movements that occur at the hip joint
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1294	Describe the hip joint and its cartilage and ligamentous support
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1295	Describe the blood supply of the hip joint and the clinical implications of it changing with age.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1297	Describe the clinical significance of an intracapsular versus an extracapsular fracture of the femur and identify on imaging studies.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1296	Describe the cruciate and trochanteric anastomoses and their clinical importance.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1298	Describe the general attachments of the muscles of gluteal region, their actions and neurovascular supply.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1299	Explain how the hip abductor muscles function during the different stages of gait
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1300	Describe the appearance of Trendelenburg sign and its causes
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1301	Describe the cutaneous innervation of the gluteal region
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1302	Describe nerves at risk during intramuscular injections in the gluteal region
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1303	Describe the general attachments of the fascia lata and the iliotibial tract (band).
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1304	1. List the muscle compartments of the thigh. 2. Describe the general attachments of the muscles of thigh region, their actions and neurovascular supply.
Lecture 16	ANAT	Clinical Anatomy of the Lower Limb - Thigh, Knee Joint, Leg	SOM.MK.II.BPM1.1.MSK.1.ANAT.1305	Describe the structures in the femoral sheath and their relationship to one another.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1306	List the boundaries and the contents of the femoral triangle, adductor canal and popliteal fossa.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1307	Describe the course of a femoral hernia and how it differs from an inguinal hernia.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1308	Discuss the consequences of piriformis syndrome, paying particular attention to the course of sciatic nerve through the piriformis muscle (SG Article)
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1310	Describe the movements that occur at the knee joint
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1311	Describe the knee joint its cartilage and ligamentous support
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1312	Describe Genu Valgus and Varus
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1313	Describe the function of the anterior and posterior cruciate ligaments.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1314	Describe the anterior and posterior drawer sign and the ligaments tested during each.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1315	Describe the function of the collateral ligaments and menisci of the knee joint
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1316	Describe the "unhappy triad" in knee joint injury.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1317	Describe the significance of the prepatellar, suprapatellar and infrapatellar bursae.

			SOM.MK.II.BPM1.1.MSK.1.ANAT.1318	Describe the blood supply of the knee joint.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1319	Describe the crural fascia.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1320	List the muscle compartments of the leg and describe their general attachments, actions and neurovascular supply.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1321	Explain the consequence of a lesion of the tibial nerve in the popliteal fossa.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1322	Explain the consequence of a lesion of the common fibular nerve at the neck of the fibula.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1323	Explain the consequence of a lesion of the superficial and deep fibular nerves just distal to their origin.
DLA 07	ANAT	Clinical Anatomy of the Lower Limb - Ankle, Foot, Arterial Supply, Venous and Lymphatic Drainage	SOM.MK.II.BPM1.1.MSK.1.ANAT.1325	Describe the movements that occur at the ankle joint
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1326	Describe the ankle joint its and ligamentous support
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1327	List the two ligaments most commonly damaged during an inversion sprain.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1328	Describe the medial tarsal tunnel and its contents.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1329	Describe the general arrangement between the tarsal bones and the metatarsals
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1330	Describe hallux valgus and hallux varus.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1331	Describe the movements that occur at the tarsometatarsal, metatarsophalangeal and interphalangeal joints of the big toe and of digits 2-5.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1332	Describe the general attachments of the muscles of sole of the foot, their actions and neurovascular supply.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1333	Describe the cutaneous innervation of the thigh, leg and foot.
Lecture 17	ANAT	Clinical Imaging of the Lower Limb	SOM.MK.II.BPM1.1.MSK.1.ANAT.1334	Predict the regional lymph nodes which will be most likely enlarged because of infection(s) in the lower limb.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1504	Describe the arterial supply, venous and lymphatic drainage of the lower limb.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1336	Identify the structures of the hip joint, femur, knee joint, tibia, fibula, ankle joint and foot in imaging studies use this information to interpret clinical signs and symptoms.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1337	Describe the posterior dislocation of the hip and the nerve most likely to be affected by this injury.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1338	Identify posterior dislocation of the hip joint on a radiograph.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1297	Describe the clinical significance of an intracapsular versus an extracapsular fracture of the femur and identify on imaging studies.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1496	Identify femoral fractures in imaging studies and discuss likely clinical signs as symptoms as a result of these fractures
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1497	Identify injuries to the ligaments and cartilages of the knee on imaging studies
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1339	Describe the patellar tendon reflex including the nerve roots being tested
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1340	Describe a "Baker's cyst" and explain possible sequelae.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1498	Identify fractures tibial and fibular fractures in imaging studies and discuss likely clinical signs as symptoms as a result of these fractures
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1324	Describe the consequences and clinical presentation of "a compartment syndrome"
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1499	Describe signs and symptoms resulting from various areas of blockage of the blood supply of the lower limb
DLA 08	ANAT	Limb Development	SOM.MK.II.BPM1.1.MSK.1.ANAT.1500	Identify the structures of the gluteal region, compartments of the thigh and leg in imaging studies and use this information to interpret clinical signs and symptoms.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1335	Describe the development of varicose veins.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1341	Describe the Achilles tendon reflex including the nerve roots being tested.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1343	Describe the musculoskeletal development of the limbs from the lateral intraembryonic mesoderm and myotomes of the somites.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1344	Asses the importance of sonic hedgehog, fibroblast growth factor and WNT in the development of the limbs
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1345	Explain the formation of the following: congenital talipes (clubfoot), amelia / meromelia, cutaneous syndactyly, osseous syndactyly, brachydactyly, polydactyly, ectrodactyly (cleft hand or partial adactyly) and limb deformities caused by amniotic bands.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1346	List and explain the clinical significance of the VACTERL association
Lecture 19	HCB	Histological Classification and Organization of Bone	SOM.MK.I.BPM1.2.MSK.3.HCB.0986	Describe bone as a specialized connective tissue in terms of its cells, fibers and ground substance.
			SOM.MK.I.BPM1.2.MSK.3.HCB.0987	List the major components of the extracellular matrix of bone.
			SOM.MK.I.BPM1.2.MSK.3.HCB.0988	Discuss the significance of the major components of the extracellular matrix of bone.
			SOM.MK.I.BPM1.2.MSK.3.HCB.0989	Differentiate between the following cells of bone based on their histological structure, function and localization: osteoprogenitor cell, osteoblast, osteocyte and osteoclast.
			SOM.MK.I.BPM1.2.MSK.3.HCB.0990	Differentiate between periosteum and endosteum based on their histological structure, function and localization.
			SOM.MK.I.BPM1.2.MSK.3.HCB.0991	Compare and contrast immature bone vs. mature bone.
			SOM.MK.I.BPM1.2.MSK.3.HCB.0992	Compare and contrast spongy bone vs. compact bone in terms of location.
			SOM.MK.I.BPM1.2.MSK.3.HCB.0993	Define an "osteon"
			SOM.MK.I.BPM1.2.MSK.3.HCB.0994	Describe the normal development of an osteon.
			SOM.MK.I.BPM1.2.MSK.3.HCB.0995	Define "osteoid".
			SOM.MK.I.BPM1.2.MSK.3.HCB.0996	State the origin of osteoclasts.
			SOM.MK.I.BPM1.2.MSK.3.HCB.0997	Describe the structure of the osteoclast in relation to its function.
			SOM.MK.I.BPM1.2.MSK.3.HCB.0998	Describe the histological structures of compact bone including osteon, concentric lamellae, Haversian canal, outer & inner circumferential lamellae, interstitial lamellae and Volkmann's canal.
			SOM.MK.I.BPM1.2.MSK.3.HCB.0999	Describe the function of a canaliculi.
Lecture 20	HCB	Histology of Bone Formation, Maintenance and Repair	SOM.MK.I.BPM1.2.MSK.3.HCB.1000	Describe the process of "ground bone preparation" to study the structure of compact bone.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1001	List the two types of bone formation.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1002	Describe the process of intramembranous ossification.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1003	Describe the process of endochondral ossification.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1004	Identify trabeculae & bony spicules.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1005	State the function of the epiphyseal plate.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1006	Describe the zones of the epiphyseal plate.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1007	Describe the process of bone repair.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1008	Describe the process of bone remodeling.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1009	Describe the hormonal regulation of the bone growth and relate it with the development of gigantism, acromegaly and dwarfism.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1010	Describe the influence of the following hormones on bone development and maintenance of blood calcium levels: calcitonin, parathyroid and growth hormones.

			SOM.MK.I.BPM1.2.MSK.3.HCB.1011	Describe the main histological and clinical features of rickets.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1012	Describe the main histological and clinical features of osteomalacia.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1013	Describe the main histological and clinical features of osteoporosis
Lecture 18	HCB	Flipped Class: Histology of Cartilage	SOM.MK.I.BPM1.2.MSK.3.HCB.1014	List the 3 types of cartilage
			SOM.MK.I.BPM1.2.MSK.3.HCB.1015	Describe the histological components specific to cartilage
			SOM.MK.I.BPM1.2.MSK.3.HCB.1016	Describe the various components of extracellular matrix of cartilage in general.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1017	Describe the microanatomical features of chondroblasts.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1018	Describe the microanatomical features of chondrocytes.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1019	Give the function(s) of the perichondrium.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1020	Give the location of perichondrium.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1021	Describe the microscopic of the perichondrium.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1022	Discuss the two types of growth seen in cartilage in relation to the perichondrium and isogenous groups.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1023	Identify the location of capsular, territorial and inter-territorial matrices.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1024	Make a comparison table differentiating the three types of cartilage.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1025	State the function(s) of hyaline cartilage.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1026	Describe the histological structure of hyaline cartilage.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1027	Give examples of the localizations of hyaline cartilage.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1028	State the function(s) of articular cartilage
			SOM.MK.I.BPM1.2.MSK.3.HCB.1029	Describe the histological structure of articular cartilage.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1030	Give examples of locations of articular cartilages
			SOM.MK.I.BPM1.2.MSK.3.HCB.1031	State the function of elastic cartilage.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1032	Describe the microanatomical structure of elastic cartilage.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1033	Give examples of locations of elastic cartilage.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1034	State the function of fibrocartilage.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1035	Describe the microanatomical structure of fibrocartilage.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1036	Give examples of locations of fibrocartilage.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1037	Discuss osteoarthritis and relate the histological changes in cartilage with clinical presentation.
DLA 9	BCHM	Vitamin D	SOM.MK.III.BPM1.2.MSK.1.BCHM.0200	Describe vitamin D formation in the skin and its conversion to calcitriol (in liver and kidney) indicating factors affecting this process.
			SOM.MK.III.BPM1.2.MSK.1.BCHM.0201	Describe the functions of 1,25 dihydroxy-D (calcitriol) related to calcium and phosphate metabolism
			SOM.MK.III.BPM1.2.MSK.1.BCHM.0202	Indicate vitamins that act as steroid hormones (calcitriol)
			SOM.MK.III.BPM1.2.MSK.1.BCHM.0203	Indicate clinical features and manifestations of hypervitaminosis D
			SOM.MK.III.BPM1.2.MSK.1.BCHM.0204	Describe the basis of clinical manifestation in rickets and osteomalacia. Identify risk factors for development of these disorders.
Lecture 21	BCHM	Calcium, Phosphate, Vitamin D	SOM.MK.I.BPM1.2.MSK.1.BCHM.0205	Indicate the nutritional needs as it relates to calcium, phosphate and vitamin D
			SOM.MK.I.BPM1.2.MSK.1.BCHM.0206	Describe the role of calcium, phosphate and vitamin D in nutrition and in the development of bone disease
			SOM.MK.I.BPM1.2.MSK.1.BCHM.0207	Describe the various functions of calcium and phosphate in humans
			SOM.MK.I.BPM1.2.MSK.1.BCHM.0208	Identify the three forms of extracellular calcium, and their contribution to total extracellular calcium
			SOM.MK.I.BPM1.2.MSK.1.BCHM.0209	Describe hypocalcemia and hypercalcemia as well as their causes, manifestations, diagnosis and treatment
			SOM.MK.I.BPM1.2.MSK.1.BCHM.0211	Describe the associations among intestinal absorption, renal elimination, bone stores, and the function of calcitriol and parathyroid hormone in regulating serum calcium and phosphate
			SOM.MK.I.BPM1.2.MSK.1.BCHM.0212	Correlate the skeletal and non-skeletal actions of vitamin D; reasons for and likely extent of its deficiency; likely deficiency diseases and potential solutions in terms of food fortification and supplementation.
			SOM.MK.I.BPM1.2.MSK.1.BCHM.0213	Describe lab findings (elevated alkaline phosphatase) in osteomalacia and rickets.
			SOM.MK.I.BPM1.2.MSK.1.BCHM.0214	Correlate hereditary resistance to vitamin D (vitamin D resistant rickets) to the biochemical mechanisms
DLA 10	GNET	Genetic Cases: OI and DMD	SOM.MK.III.BPM1.2.MSK.1.GNET.0156	Discuss elevated creatine kinase levels in DMD and BMD patients
			SOM.MK.III.BPM1.2.MSK.1.GNET.0157	Describe the inheritance pattern of Duchenne muscular dystrophy and discuss the clinical features
			SOM.MK.III.BPM1.2.MSK.1.GNET.0158	Compare and contrast the allelic heterogeneity in the DMD gene that leads to either Becker muscular dystrophy or Duchenne muscular dystrophy
			SOM.MK.III.BPM1.2.MSK.1.GNET.0159	Analyze the DMD gene and discuss why it is sometimes a hotspot for mutation in the human genome
			SOM.MK.III.BPM1.2.MSK.1.GNET.0160	Discuss de novo mutations which lead to 1/3 of all cases of DMD
			SOM.MK.III.BPM1.2.MSK.1.GNET.0161	Explain the effects of skewed X-inactivation leading to a manifesting heterozygote in females
			SOM.MK.III.BPM1.2.MSK.1.GNET.0162	Describe the structure of collagen
			SOM.MK.III.BPM1.2.MSK.1.GNET.0163	Describe the general features of osteogenesis imperfecta
			SOM.MK.III.BPM1.2.MSK.1.GNET.0164	Describe how classic non-deforming OI with blue sclera (formerly known as OI type 1) using the haploinsufficiency model.
			SOM.MK.III.BPM1.2.MSK.1.GNET.0165	Describe the perinatally lethal osteogenesis imperfecta (formerly known as OI type 2) using the dominant negative model.
Lecture 22	GNET	Flipped Class: Inherited Disorders of the Musculoskeletal System	SOM.MK.III.BPM1.2.MSK.1.GNET.0154	Review the genetic mechanisms in DMD, OI, NF-1, CMT, Marfan Syndrome, Achondroplasia and Alzheimer disease
Lecture 23	PHYS	Neuromuscular Transmission and Excitation-Contraction Coupling	SOM.MK.II.BPM1.2.MSK.2.PHYS.0052	Explain the overall transmembrane signaling steps whereby increases in cytosolic calcium initiate crossbridge cycling
			SOM.MK.II.BPM1.2.MSK.2.PHYS.0053	Identify the multiple sources, localization, and roles of calcium in muscle contraction and relaxation.
			SOM.MK.II.BPM1.2.MSK.2.PHYS.0054	Illustrate the chemical and mechanical steps in the cross bridge cycle, and explain how the cross bridge cycle results in shortening of the muscle.
			SOM.MK.II.BPM1.2.MSK.2.PHYS.0055	Explain the relationship of the myosin-thick filament bare zone (I zone) to the shape of the active length:force relationship.
			SOM.MK.II.BPM1.2.MSK.2.PHYS.0056	List the steps in excitation contraction coupling in skeletal muscle, and describe the roles of the sarcolemma, transverse tubules, sarcoplasmic reticulum, thin filaments, and calcium ions.
			SOM.MK.II.BPM1.2.MSK.2.PHYS.0057	Describe the roles of ATP in skeletal muscle contraction and relaxation.
			SOM.MK.II.BPM1.2.MSK.2.PHYS.0058	List in sequence the steps involved in neuromuscular transmission in skeletal muscle and point out the location of each step on a diagram of the neuromuscular junction.

			SOM.MK.II.BPM1.2.MSK.2.PHYS.0059	Distinguish between an endplate potential and an action potential in skeletal muscle.
			SOM.MK.II.BPM1.2.MSK.2.PHYS.0060	List the possible sites for blocking neuromuscular transmission in skeletal muscle and provide an example of an agent that could cause blockage at each site.
			SOM.MK.II.BPM1.2.MSK.2.PHYS.0061	Describe the role of the myosin crossbridges acting in parallel to determine active force.
Lecture 24	PHYS	Flipped Class: Mechanics of Skeletal Muscle Contraction	SOM.MK.II.BPM1.2.MSK.2.PHYS.0062	Compare and contrast between an isometric and isotonic contraction.
			SOM.MK.II.BPM1.2.MSK.2.PHYS.0063	Illustrate the length versus force diagram for muscle and label the three lines that represent passive (resting), active, and total force. Explain the molecular origin of these forces in the three muscle types.
			SOM.MK.II.BPM1.2.MSK.2.PHYS.0064	Explain the interaction of the length:force and the force:velocity relationships and how they vary in the three muscle types.
			SOM.MK.II.BPM1.2.MSK.2.PHYS.0065	Distinguish between a twitch and tetanus in skeletal muscle and explain why a twitch is smaller in amplitude than tetanus and the continuum of force development between a twitch and tetanus including the intracellular events.
			SOM.MK.II.BPM1.2.MSK.2.PHYS.0066	Illustrate force versus velocity relationships for two skeletal muscles of equal maximum force generating capacity but of different maximum velocities of shortening.
			SOM.MK.II.BPM1.2.MSK.2.PHYS.0077	Identify the energy sources for muscle contraction. Categorize the three muscle types on the basis of energy sources, their relative speed of contraction, and capacity to supply ATP for contraction.
			SOM.MK.II.BPM1.2.MSK.2.PHYS.0078	Explain the basis of muscular fatigue. List factors that can cause fatigue.
Lecture 25	PHYS	Physiology of Cardiac and Smooth Muscles	SOM.MK.II.BPM1.2.MSK.2.PHYS.0067	Contrast the duration of the action potential and the refractory period in a cardiac muscle, a skeletal muscle, and a nerve. Sketch the temporal relationship between an action potential in a cardiac muscle cell and the resulting contraction (twitch) of that cell. On the basis of that graph, explain why cardiac muscle cannot remain in a state of sustained (tetanic) contraction.
			SOM.MK.II.BPM1.2.MSK.2.PHYS.0068	List the steps in excitation-contraction coupling in the cardiac myocytes from the initiation of an action potential to the relaxation of the cell
			SOM.MK.II.BPM1.2.MSK.2.PHYS.0069	Provide specific details about the special role of Ca^{2+} in the control of contraction and relaxation of cardiac muscle.
			SOM.MK.II.BPM1.2.MSK.2.PHYS.0070	Compare cardiac and skeletal muscle with respect to: cell size, electrical connections between cells, and arrangement of myofilaments. Based on ion permeability and electrical resistance describe role of gap junctions in creating a functional syncytium.
			SOM.MK.II.BPM1.2.MSK.2.PHYS.0071	Identify the role of extracellular calcium in cardiac muscle contraction. Identify other sources of calcium that mediate excitation-contraction coupling, and describe how intracellular calcium concentration modulates the strength of cardiac muscle contraction (Inotropism).
			SOM.MK.II.BPM1.2.MSK.2.PHYS.0072	Compare the similarities and differences in the mechanisms of actomyosin regulation in the smooth muscle vs striated muscle (cardiac and skeletal muscles)
			SOM.MK.II.BPM1.2.MSK.2.PHYS.0073	Explain the concept of latch mechanism in the smooth muscle cells, and how these cells can develop and maintain force with a much lower rate of ATP hydrolysis than skeletal muscle
			SOM.MK.II.BPM1.2.MSK.2.PHYS.0074	List the intracellular pathways that control the contraction and relaxation in smooth muscle cells and explain the differences between electromechanical and pharmacomechanical coupling
			SOM.MK.II.BPM1.2.MSK.2.PHYS.0075	Compare the characteristics of the multi-unit and unitary smooth muscle types
			SOM.MK.II.BPM1.2.MSK.2.PHYS.0076	Explain the myofilament calcium sensitization and desensitization mechanisms
DLA 11	GNET	Genetic Cases: Triplet Repeats and Pharmacogenetics	SOM.MK.III.BPM1.2.MSK.1.GNET.0166	Explain how repetitive sequences show instability such that in some individuals the size of the repeat expands in the germ line
			SOM.MK.III.BPM1.2.MSK.1.GNET.0167	Explain how most triple repeat disorders show decreased age of onset and increased severity when the repeat size expands and how parental origin may sometimes influence expansions and therefore anticipation
			SOM.MK.III.BPM1.2.MSK.1.GNET.0168	Describe fragile X syndrome
			SOM.MK.III.BPM1.2.MSK.1.GNET.0169	Explain how fragile X syndrome is a disorder caused by a loss of expression of a gene
			SOM.MK.III.BPM1.2.MSK.1.GNET.0170	Describe fragile X-associated tremor ataxia syndrome and how it is caused by an overexpression of a gene
			SOM.MK.III.BPM1.2.MSK.1.GNET.0171	Describe genetic testing and therapies (antisense oligonucleotides) for Huntington Disease
			SOM.MK.III.BPM1.2.MSK.1.GNET.0172	Describe the genetics and clinical features of myotonic dystrophy
			SOM.MK.III.BPM1.2.MSK.1.GNET.0173	Describe the genetics and clinical features of Friedrich ataxia and spinocerebellar ataxia
			SOM.MK.III.BPM1.2.MSK.1.GNET.0174	Outline the classic examples of gene-drug interactions
			SOM.MK.III.BPM1.2.MSK.1.GNET.0175	Compare and contrast the terms pharmacogenetics and pharmacogenomics
			SOM.MK.III.BPM1.2.MSK.1.GNET.0176	Explain the pathophysiology of malignant hyperthermia
			SOM.MK.III.BPM1.2.MSK.1.GNET.0177	Analyze the genetic aspects of malignant hyperthermia including autosomal dominant inheritance with reduced penetrance as well as locus vs. allelic heterogeneity
			SOM.MK.III.BPM1.2.MSK.1.GNET.0178	Describe the cytochrome p450 family of enzymes and their general function
			SOM.MK.III.BPM1.2.MSK.1.GNET.0179	Discuss the CYP2D6 enzyme and how it metabolizes metoprolol and codeine
Lecture 26	GNET	Flipped Class: Triplet Repeat Disorders and Pharmacogenetics	SOM.MK.III.BPM1.2.MSK.1.GNET.0174	Outline the classic examples of gene-drug interactions
			SOM.MK.III.BPM1.2.MSK.1.GNET.0155	Analyze the genetic principles of triple repeat disorders.
Lecture 27	HCB	Histology of the Skin	SOM.MK.I.BPM1.2.MSK.3.HCB.1054	Describe the 5 layers of the epidermis and its associated cell types.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1055	Describe cytokeratins.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1056	Describe keratohyalin granules.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1057	Describe the steps in epidermal cell differentiation, and cell renewal in relation to the epidermal layers.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1058	Describe the steps in the process of keratinization.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1059	State the function of the epidermal water barrier.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1060	Describe the microanatomical structure of the epidermal water barrier.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1061	Compare and contrast the typical epidermal turnover time vs. the hyperproliferative disease psoriasis.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1062	List at least 3 factors associated with skin pigmentation.

			SOM.MK.I.BPM1.2.MSK.3.HCB.1063	Differentiate between thick and thin skin based on location and histological features.
DLA 12	HCB	An Overview of the Histopathological Aspects of Skin	SOM.MK.I.BPM1.2.MSK.3.HCB.1043	Identify the cell responsible for the development of melanoma and epidermal features listed in the diagnostic criteria.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1044	Discuss the 3 classifications of burns.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1045	List the 4 stages of wound healing.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1046	Describe the changes to the skin that results in pemphigus vulgaris.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1047	Describe skin pigmentation changes seen in vitiligo.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1048	Describe skin pigmentation changes seen in albinism.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1049	Describe the inflammatory changes seen in sebaceous glands in the clinical condition called acne.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1050	Describe the changes to the skin that results in bullous pemphigoid.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1051	Discuss keloid formation.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1052	Define acanthosis, acantholysis, hyperkeratosis and parakeratosis.
Lecture 28	HCB	Histology of the Skin Appendages	SOM.MK.I.BPM1.2.MSK.3.HCB.1064	Describe the layers of the dermis.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1065	Identify the epidermal skin appendages (hair, nails, sebaceous glands, sweat glands).
			SOM.MK.I.BPM1.2.MSK.3.HCB.1066	Give the function(s) of hair follicles.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1067	Describe the histological structure of hair follicles.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1068	Describe the stages of hair growth
			SOM.MK.I.BPM1.2.MSK.3.HCB.1069	Describe the microanatomical structure of finger nails.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1070	Describe the microanatomical structure of toe nails.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1071	Describe the parts responsible for nail growth.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1072	Give the function(s) of sebaceous glands.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1073	State the localization of sebaceous glands.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1074	Describe the microscopic structure of sebaceous glands.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1075	Give the function(s) of eccrine sweat glands.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1076	State the localization of eccrine sweat glands.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1077	Describe the microscopic structure of eccrine sweat glands.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1078	Give the function(s) of apocrine sweat glands.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1079	State the localization of apocrine sweat glands.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1080	Describe the microscopic structure of apocrine sweat glands.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1081	Compare and contrast the mode of secretion in sebaceous glands vs. eccrine sweat glands vs. apocrine sweat glands.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1082	List the function(s) of free nerve endings in the epidermis.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1083	Describe the microscopic structure of free nerve endings in the epidermis.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1084	List the function(s) of Pacinian corpuscles.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1085	State the localization of Pacinian corpuscles.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1086	Describe the microscopic structure of Pacinian corpuscles.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1087	List the function(s) of Meissner's corpuscles.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1088	State the localization of Meissner's corpuscles.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1089	Describe the microscopic structure of Meissner's corpuscles.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1090	List the function(s) of Ruffini's corpuscles.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1091	State the localization of Ruffini's corpuscles.
			SOM.MK.I.BPM1.2.MSK.3.HCB.1092	Describe the microscopic structure of Ruffini's corpuscles.